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Measuring nurse managers' boundary spanning: development and psychometric evaluation

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Aim To test the psychometric properties of a boundary spanning measure by nurse managers.

Background The health-care environment requires hospital units to coordinate efforts autonomously across their boundaries and to manage relationships with other professionals, units and departments. Boundary spanning has become increasingly important for first-line nurse managers as unit gatekeepers; however, the available measures are limited.

Method The 30-item instrument was developed from a literature review. Survey participants were 4918 nurses at 231 hospital units. Statistical analyses of construct validity and internal consistency were performed. Furthermore, the correlation between nurses' scores on the Nurse Managers Boundary Spanning Scale and nurses' evaluations of their managers were examined.

Results Three factors and 26 items were derived from factor analyses: connecting and mediating, informing and feedback utilisation, and resource acquisition. Cronbach's subscales' alpha coefficients were above 0.9. Correlation analysis indicated that the Nurse Managers Boundary Spanning Scale score correlated with nurses' positive perceptions of their managers.

Conclusion This study demonstrates tentative support for the validity and reliability of the Nurse Managers Boundary Spanning Scale. Although further study is needed, the Nurse Managers Boundary Spanning Scale shows possibilities as a new measurement of nursing leadership.

Implications This study underscores measures to build on nurse managers' roles by building on the limited research available.

Keywords: boundary spanning, nurse managers, Nurse Managers Boundary Spanning Scale, staff perception

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Introduction

Nurses' work in hospital units relies on various inputs from outside the unit, such as information on patients' needs and new technologies, support from other professionals, departments and upper management. Nurse leaders must create a positive work environment by ensuring access to relevant resources and information, enhancing collaborative relationships between physicians and nurses, and linking staff nurses to hospital-wide issues (Heale *et al.* 2014, Macphee *et al.* 2014, Stam *et al.* 2015). In particular, first-line nurse managers are gatekeepers in their units (Lewis 1991) and are the vital link between senior management and

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staff nurses (Marx 2014). Efforts to coordinate input flows, such as information and resources, to negotiate the necessary support and to manage a unit's relationships with upper management or other units are referred to as boundary spanning (Ancona & Caldwell 1992, Marrone 2010). As Meyer *et al.* (2011) noted, first-line managers are typically boundary spanners. Although boundary spanning is one of the nurse managers' roles, there has been limited research focusing on managers' boundary spanning and its outcomes for their units. Accordingly, this study was designed to develop and test a measure of nurse managers' boundary spanning behaviour.

Background

Organisation management literature emphasises that changing environments and flatter organisational structures require organisational work groups to coordinate work efforts across boundaries and to manage relationships with external groups or organisations (Faraj & Yan 2009, Marrone 2010). Based on the open-system theory, Ancona and Caldwell (1988) introduced external perspectives into work group research. They indicated that existing research tends to focus only on behaviour within groups, such as classifying group functions by task and maintenance activities, regardless of the work groups' interdependence with other organisational units, and they argued the importance of the groups' interactions with external parties. The importance of such activities for information transmission has been noted in the study of a research and development facility (Tushman 1977). Gladstein (1984) also found that sales team members perceived the management of interactions across their boundaries as separate and distinct from internal activities, such as task and maintenance behaviours. Ancona (1990) and Ancona and Caldwell (1988, 1992) conducted studies on new-product teams to identify and classify their external activities, i.e. boundary spanning. Subsequent researchers have noted that today's knowledge-oriented organisations require the effective transfer of non-codified and complex forms of knowledge across organisational units, the coordination of efforts and the external sharing of resources. Based on studies of multiple types of work groups or teams, they have also demonstrated that boundary spanning in work groups is associated with innovation (Tsai 2001, Grawe *et al.* 2015), performance (Faraj & Yan 2009), effectiveness (Marrone *et al.* 2007) and team learning (Edmondson & Nembhard 2009).

Marrone (2010, p. 914), following a review of the organisation-management literature, defined team boundary spanning as 'the team's actions to establish linkages and manage interactions with parties in the external environment'. In addition, Marrone (2010) has demonstrated the taxonomy of boundary-spanning actions. The taxonomy comprises three behavioural categories: (1) representation, (2) coordination of task performance and (3) general information search.

Representation reflects interactions mainly directed toward others who have greater power than the team and includes actions that provide information about the team's activities, obtain understanding of the team's expectations, persuade others of the team's decisions and protect the team from overloading and pressure. Representation enables the team to establish a mutual understanding with upper management, obtain resources for work and devote the team's effort to achieve its goal. Coordination of task performance reflects interaction with the team's lateral external groups and includes actions that confirm the progress of a shared task, coordinate each effort and obtain feedback. Coordination of task performance enables the team to establish mutual support and cooperation with outside staff so that the efficiency and effectiveness of work is enhanced. General information search involves actions that access outside parties for general or technical information or expertise; it also enables the team to perceive environmental change related to their work and provides them with opportunities for learning and innovation.

In actions reflected by representation and coordination of task performance, the team leader is often considered a central performer. Ancona and Caldwell (1990) found that the team leader's external strategy influenced the interaction between the team and its environment and team performance. Richter *et al.* (2006) also deemed team leaders to be the primary conduits for both the coordination of intergroup activities and the negotiation of intergroup conflicts in the organisations, and demonstrated that a team leader's group identification, organisational identification and frequency of intergroup contact relate to effective intergroup relations.

Boundary spanning roles in nursing organisations

Nurses in hospital units are one kind of work group, and it is not difficult to understand the importance of boundary spanners for them. Previous studies demonstrated the importance of some aspects of boundary spanning. Structural empowerment, which is reflected by nurses' perceptions of access to opportunity, information, support and resources, is significantly correlated with patient-safety culture (Armellino *et al.* 2010), and fosters the nurses' perceived unit support for professional nursing practice and perceptions of unit effectiveness (Laschinger *et al.* 2014). Sheingold and Sheingold (2013) proposed that the framework of social capital, which represents the quality of relationships and networks available to individuals within communities or organisations, was useful for measuring the nursing work environment. From a team learning perspective, Edmondson (2003) found that operating-room teams need external communication to coordinate their efforts with others outside team boundaries to introduce new technologies. Although existing frameworks or measures are useful for managing health-care organisations or units, boundary spanning focuses more on relationship management roles, including filtering information from the environment to prevent organisation members or influencing parties outside the organisation from accommodating the preferences of their organisation. For hospital unit nurses, a relationship management perspective is useful because of their need to negotiate with upper management or other professionals to improve the care environment. In addition, boundary spanning includes acquiring feedback on the unit's work progression. Feedback on daily practice can contribute to adherence to practice guidelines and the quality of care (Dulko & Mooney 2010, Kaasalainen *et al.* 2015). Duffield *et al.* (2011) revealed that staff nurses perceived nurse managers who provide positive feedback and consult with staff on daily problems as good leaders.

Defining nurse managers' boundary spanning

Based on Marrone's (2010) review, nurse managers' boundary spanning was defined in this study as their actions to establish linkages and to manage interactions between their units and parties inside and outside the hospital, such as other professionals, other units, other organisations and patients. According to Marrone's (2010) taxonomy of boundary spanning and leaders' roles indicated by previous studies, the concept of nurse managers' boundary spanning assumes two constructs: the representation and the coordination of task performance.

Health-care facilities involve several teams, including the nursing team and a multi-professional team. Nurse managers must manage both. In this study, however, we focus on nurse managers in their capacity as representatives of nursing teams. Generally, large acute care hospitals in Japan each consist of specialised departments such as the medical department, nursing department and rehabilitation department. Although they collaborate in patient care, they must also manage their own affairs. In nursing departments, boundaries between upper management and individual nursing teams and across nursing teams as well as boundaries across professionals, should be managed to establish standards for nursing and common goals for patient care.

The study

Aim

The aim of the study was to test the validity and reliability of the measure of nurse managers' boundary spanning [the Nurse Managers Boundary Spanning Scale (NMBS)].

Methodology

The NMBS was developed as a part of a larger study to investigate factors enhancing the effectiveness of a nursing team. The NMBS was devised by (1) item generation and refinement by content validation and pilot study and (2) testing for reliability and validity with a cross-sectional study. Data were collected from July to August 2011.

Instrument

Item generation and refinement

The items of the NMBS were developed based on a review of the organisation-management literature related to boundary spanning and its existing instruments (e.g. Ancona & Caldwell 1992, Druskat & Wheeler 2003, Hirst & Mann 2004, Faraj & Yan 2009). These items were modified by the authors for better applicability to practical hospital settings. We assumed two constructs for managers' boundary spanning (the representation and the coordination of task performance); however, the literature review indicated that each of these constructs included a broad range of activities and could be further subcategorised. Therefore, we categorised six hypothetical conceptual domains of nurse managers' boundary spanning. Representation was broken into four dimensions: (1) informing and persuading, which reflects informing others of units' goals and activities, their importance, and their progress; and persuading others to obtain external support as needed; (2) buffering, which represents controlling outside pressures and requests to protect team members from conflict and overloading; (3) clarifying the position, which reflects communicating to team members policies, decisions, evaluations and expectations about hospital and nursing divisions and enhancing their understanding of these; and (4) connecting, or understanding where unit nurses can obtain support and resources and supporting unit members' access to these. The coordination of task performance was grouped into two categories: (5) feedback utilisation, which addresses obtaining and bringing in feedback from other professionals about a unit's performance; and (6) cooperation and coordination, or actions to obtain cooperation and coordinate efforts with other units and professionals to improving the unit's performance. The generated items were assigned to these domains based on these definitions.

Following the generation of items, we held discussions with researchers in nursing administration and nurse administrators to assess the content's validity and the items' suitability, as well as to assess nurse manager behaviour in a practical hospital setting. We also solicited the opinions of nurse managers and nurses working at the hospitals. After the wording of a few items was modified based on their input, they deemed all items relevant and appropriate.

The NMBS is a scale to assess nurse managers' behaviour, and an issue raised in the process of designing the scale was who should assess it. We designed the NMBS to assess staff perceptions of their nurse managers' behaviour because it has the advantage of greater objectivity than self-assessment. In addition, the nurse managers' boundary spanning benefits unit nurses daily and brings them useful information and resources. Therefore, the NMBS is designed to ask nurses the extent to which their managers perform each relevant behaviour, and each item is rated on a 5-point Likert-type scale ranging from 1 (**never**) to 5 (**very great extent**). A higher score indicates that the nurse perceives that his/her nurse manager performs boundary spanning well. As a result, a 30-item instrument was developed as the NMBS and used in a pilot study.

The pilot study was conducted with 69 staff nurses who worked at an acute care hospital. Data obtained from the participants indicated no ceiling and no floor effects for each item, and no changes were made as a result of the pilot study.

NMBS reliability and validity

Participants

To test for reliability and validity, 29 acute care hospitals in Japan were contacted and consented to participate in this study. Each hospital had over 200 beds. Questionnaires were distributed to 5809 nurses in 231 units, excluding the nurse managers working at those hospitals. Questionnaire recipients were asked to complete the NMBS and return it via the mailbox placed in each unit.

In addition, to examine whether the NMBS is a useful indicator of nursing management, we recruited 166 nurses at an acute-care hospital as another sample. They were asked to respond to the NMBS and an additional item evaluating their nurse manager as a unit leader: 'The nurse manager is a good leader for nurses in this unit'. The response was rated on a five-point Likert-type scale ranging from 1 (**strongly disagree**) to 5 (**strongly agree**). Higher scores indicate that nurses evaluated their manager positively.

Data analysis

Exploratory factor analysis was conducted to assess construct validity. We used the maximum likelihood extraction with a promax rotation. The number of factors was established based on eigenvalues over 1.0. The internal consistency of the whole scale and subscales was evaluated using Cronbach's alpha. Subscales were considered acceptable as a consistent measure if Cronbach's $\alpha \geqq 0.70$.

In addition, to examine criterion validity, Pearson's correlation coefficients were calculated for the NMBS with nurses' evaluation of the manager that 166 nurses answered.

Table 1
Participants' demographic characteristics (n = 4918)

	n (%)
Gender	
Male	289 (6.0)
Female	4434 (92.8)
Age ($M \pm$ SD)	32.2 ± 8.5
Years of service ($M \pm$ SD)	8.9 ± 8.3
Years of hospital service ($M \pm$ SD)	6.9 ± 7.6
Years of unit service ($M \pm$ SD)	2.5 ± 3.0
Highest degree	
Assistant nurse diploma	174 (3.6)
Diploma	3256 (68.0)
Three-year college (Associate's degree)	521 (10.9)
College (Bachelor's degree)	740 (15.5)
Graduate school	33 (0.7)

Statistical analyses were conducted with IBM SPSS Statistics 19.0 (IBM SPSS, Tokyo, Japan) software.

Ethical considerations

The study was approved by the university's ethics review board. A letter of invitation outlining the

Table 2
Explanatory factor analysis of the NMBMS (n = 4918)

Item		Hypothesised subscale	Factor loadings: Factor 1	Factor 2	Factor 3	Alpha
Connecting and mediating						
23	When problems arise between their staff members and other professions or units, the nurse manager mediates between those involved	Buffering	0.874	0.164	0.105	0.93
24	The nurse manager protects his/her staff from criticism or unreasonable demands	Buffering	0.871	0.161	0.125	
22	In the event of negative feedback on a nurse's work, the nurse manager does not simply reprimand the nurse involved, but discusses how improvements can be made	Buffering	0.802	0.056	0.083	
25	The nurse manager coordinates or negotiates as required to ensure that the unit nurses is not overloaded with work	Buffering	0.660	0.077	0.261	
3	The nurse manager explains nursing decisions and hospital administrator decisions in a way that staff can understand	Clarifying the position	0.594	0.451	0.260	
12	The nurse manager understands problems felt by staff members and consults with individuals or organisations that can provide solutions	Connecting	0.591	0.286	0.002	
16	In their role, the nurse manager identifies staff characteristics and abilities, and holds committee and review meetings within the hospital	Connecting	0.585	0.114	0.097	
18	The nurse manager passes on good appraisals from patients to staff members	Feedback utilisation	0.580	0.094	0.054	
2	The nurse manager quickly ascertains and manages policies and decisions made by nursing and hospital administrators	Clarifying the position	0.562	0.436	0.246	
20	The nurse manager listens to the thoughts and opinions of patients regarding nursing care and passes this information onto staff members	Feedback utilisation	0.545	0.142	0.129	
14	The nurse manager advises staff members where they can obtain the knowledge required to solve problems	Connecting	0.534	0.319	0.039	
15	The nurse manager supports cooperation between his/her staff and other professions and units	Connecting	0.502	0.152	0.196	
13	The nurse manager coordinates with individuals who have the knowledge to achieve the goals and solve any problems faced by the staff members	Cooperation and coordination	0.486	0.380	0.040	
Informing and feedback utilisation						
7	The nurse manager informs other professionals and units about the goals and current undertakings of his/her unit	Informing and persuading	−0.135	0.827	0.175	0.93
5	The nurse manager demonstrates what is expected of the unit by the hospital	Clarifying the position	−0.003	0.786	0.037	
6	The nurse manager receives feedback from hospital and nursing administrators on the state of work at the unit	Informing and persuading	0.070	0.766	0.029	
4	The nurse manager informs staff members how hospital administrators and the nursing department assess the work at their unit	Clarifying the position	0.097	0.732	−0.016	
8	The nurse manager persuades staff from other professions to support the goals and undertakings of the unit	Informing and persuading	0.054	0.687	0.136	
9	The nurse manager actively promotes the work outcomes of the unit to other departments	Informing and persuading	−0.035	0.683	0.217	
11	The nurse manager creates opportunities for staff members to present activities undertaken by the unit at workshops and seminars held outside the hospital	Informing and persuading	−0.069	0.520	0.276	
10	The nurse manager creates opportunities for staff members to present activities undertaken by the unit at workshops and seminars held by the hospital	Informing and persuading	0.009	0.463	0.260	
Resource acquisition						
28	The nurse manager obtains cooperation from specialists (both inside and outside of the hospital) who have expert knowledge to solve problems	Cooperation and coordination	0.051	0.087	0.767	0.95
30	The nurse manager solves problems through shared learning and problem solving with other units	Cooperation and coordination	0.045	0.155	0.720	
29	The nurse manager introduces specialists, workshops and seminars both inside and outside of the hospital that are useful for improving work	Connecting	0.028	0.128	0.697	
26	The nurse manager discusses problems related to their unit with other professionals	Cooperation and coordination	0.278	0.066	0.550	
27	The nurse manager obtains new ideas and resources required to achieve the goals and to solve the problems faced by the unit	Connecting	0.390	0.010	0.538	

Values of 0.400 or greater are highlighted.

aims and giving further details about the study accompanied each questionnaire. The questionnaires were sent to the hospital administrators and distributed to each hospital unit. Consent to participate was assumed on the basis of a returned questionnaire, and the material returned did not contain any personal information that could be used to identify the respondent.

Results

Participant demographics

Of the 5809 questionnaires distributed, 4918 questionnaires were returned (84.7% response rate). Table 1 shows participants' demographics. The majority of participants were female (92.8%), their mean age was 32, and their mean length of nursing experience was 8.9 years. Sixty-eight percent of the participants had achieved a nursing diploma as the highest level of nursing education.

Among the 231 units, 84 units were surgical units. The next most common types were internal medicine (63 units), medical/surgical (33 units) and intensive care (15 units). The mean number of staff was 25 (range 8–48).

Item analysis

The intercorrelations among items indicated that the obtained data could be applied for explanatory factor analysis. In addition, the KMO was 0.97 and Bartlett's test was significant, indicating that factor analysis was suitable for the data.

Construct validity

The factor analysis, which was not constrained by the number of possible factors, yielded three factors and explained 67.7% of the variance. We conducted another analysis constrained to a six-factor solution, which was consistent with the hypothesised subscales. However, this analysis eliminated several items that loaded on the three-factor solution, and several factors contained only two or three items. Furthermore, it was difficult to interpret each factor as a coherent unit. Thus, we adopted a three-factor solution. Four items were eliminated because they did not have a factor loading ≧0.40 (items 1, 17, 19 and 21): item 1, 'The nurse manager talks about the goals of the hospital and nursing department in a way that staff members can understand'; item 17, 'The nurse manager obtains feedback from other professionals regarding the performance of this unit'; item 19, 'The nurse manager holds discussions with staff members based on data related to the quality of nursing (degree of patient satisfaction and clinical indicators)'; and item 21, 'The nurse manager obtains advice and guidance for staff from clinical nurse specialists and certified nurses'. The factor analysis was re-run on the remaining 26 items. A summary of the three-factor solution with 26 items explaining the 68.2% of the variance as shown in Table 2.

The first factor contained all items of the hypothesised subscale 'buffering' and most items of the subscale 'connecting'. Several items from 'feedback utilisation' and 'clarifying the position' were also included. These 13 items delineated the manager's actions that helped staff connect and collaborate with external parties effectively, including buffering functions; this was labelled 'connecting and mediating'. The second factor included all items from 'informing and persuading' and two items that convey the expectations and evaluation of the unit from 'clarifying the position'; thus, this was labelled 'informing and feedback utilisation'. Although items 2 and 3 had cross-loading on both of the first and second factors, we decided to contain these items based on the value of factor-loading and their meanings. The third factor consisted of five items indicating behaviour that brings knowledge and ideas for improving unit performance; this was labelled 'resource acquisition'.

Reliability

Alpha coefficients for the three subscales were 0.93, 0.93 and 0.95, respectively; this provided support for the internal consistency reliability of the NMBS.

Correlations between the NMBS score and nurses' evaluations of their nurse managers

Correlation coefficients between the NMBS subscales and nurses' evaluations for nurse managers are presented in Table 3. All three subscales demonstrated significant positive correlation coefficients.

Table 3
Correlation between Nurse Managers Boundary Spanning Scale (NMBS) score and nurses' evaluations of their nurse manager

	The NMBS scale			
	Connecting and mediating	*Informing and feedback utilisation*	*Resource acquisition*	*Total score*
Nurses' evaluation for nurse manager	0.817*	0.699*	0.684*	0.805*

*$P < 0.01$.

Discussion

Factor structure

The NMBS is a new scale that measures nurse managers' behaviour when managing the relationships of their units with other units or groups. As a result of the exploratory factor analysis, three factors were derived: connecting and mediating, informing and feedback utilisation and resource acquisition. The NMBS measures staff perceptions of nurse managers' behaviour, and this structure shows how staff clarify their managers' behaviour. The six subscales that were hypothesised are thus roughly preserved and converged on the three factors. This might be attributed to staff nurses focusing on the benefits they may gain from nurse managers' behaviours rather than on the nurse managers' intentions. This may have been the case for items of the first factor (connecting and mediating reflected actions enabling staff to connect and collaborate with external parties) and the second factor (informing and feedback utilisation), and the third factor, which benefit staff members in terms of resource availability. Thus, the three factors share the feature of the staff nurses' potentially benefitting from their manager's actions.

Reliability and validity

The α coefficients of 0.9 and above are slightly high in relation to the desirable values (0.70–0.90) recommended by Streiner and Norman (2003). As discussed in the factor structure, participants could not discriminate between nurse managers' intentions because of the similarities of these behaviours' results. However, our results support the coherence of each factor as a unit and thus indicated a certain level of internal consistency. Further research is needed to study the possibility of a short version of the NMBS.

The criterion validity was supported by the positive correlations with nurses' evaluation of their nurse managers. The NMBS can be used as one of the indicators of a good leader. This study provided some evidence for the reliability and validity of the NMBS for assessing nurse managers' role implementation.

Implication for nursing management

The results of the present study offer preliminary support for the psychometric properties of a new instrument designed to measure the boundary spanning behaviour of nurse managers. The NMBS developed in this study described managers' behaviour to improve connections between the nursing staff and external stakeholders, including upper management, relevant professions, organisational resources and feedback information.

The importance of ensuring access to information, support and resources for nurses as means of empowering them has been noted in previous studies (Read & Laschinger 2015, Wing *et al.* 2015). However, the specific behaviours of nurse managers as they provide input on the goals of their wards have not been described. In large organisations in particular, complex organisational structures could potentially make it difficult for individual nurses to understand and adequately access the structure themselves. The NMBS developed in this study could be a guide for nurse managers to evaluate their roles as gatekeepers for their staff.

The focus of the NMBS differs in its definition of leadership from previous nursing literature. Leadership is the process of influencing another person or group in order to accomplish a goal (Huber *et al.* 2000). Although various leadership styles exist, leadership is typically represented by interpersonal skills such as motivating and inspiring subordinate nurses, sharing visions and opinions, and open and honest communication (Heuston & Wolf 2011, Wong & Giallonardo 2013). Certainly, first-line managers must possess and enact these skills; however, along with the increase in administrative management roles for first-line managers came the sharing of leadership roles among lower-ranked nurses, including staff nurses. In particular, leadership at the point of care, such as collaborating with other health professionals and encouraging colleagues for direct patient care, have come to be considered as clinical leadership and have been enacted by staff nurses as professional behaviour (Patrick *et al.* 2011). In fact, nurses other than first-line managers have exerted clinical leadership. In the USA, clinical nurse leaders provide and manage the care of individuals and cohorts of clients at the point of care by coordinating, delegating and evaluating health professionals and the care they provide (AACN 2007). In the UK, leaders who empower nurses and other health professionals to deliver quality care are not necessarily individuals in managerial positions (Stanley 2006). In Japan, a similar arrangement has recently been adopted in acute care hospitals. There are examples of systems in which nurses at assistant manager or expert levels assume leadership roles such as supervising daily nursing care, coordinating care among health professionals, and managing nursing teams (Oso *et al.* 2012). Therefore, leadership in a ward is shared, and the focus varies by position. Sharing leadership roles with other staff in the ward enable nurse managers to assume more indirect leadership roles, such as boundary spanning. The NMBS developed in this study might be a measure for new nursing leadership that reflects the changes in management roles. In the recent business literature, boundary spanning attracted attention as a new leadership skill required for managers to drive innovation and to create partnerships across boundaries (Yip *et al.* 2011). In health care, innovative practices to provide quality care are needed under the reform of health-care system and organisational restructuring in Japan and other countries. The NMBS reflects these needs for health-care administrators and managers and could be utilised as a new scale for evaluating leadership.

Study limitations

This study revealed limited parts of the validity and reliability of the NMBS; further research is needed to test the factor structure and validity of the NMBS. In addition, this study did not verify whether nurse managers' behaviour as measured by the NMBS improved the quality of the unit work. In addition, the outcomes of boundary spanning should be the quality of relationships between nursing and other professions; therefore, perspectives on nurse managers from other professions are needed. Further studies are needed to analyse the relationships between the NMBS and indicators such as quality of patient care and unit evaluation from other professionals.

In addition, this study was conducted on acute care hospitals with over 200 beds. Changing working environments for nurses and the importance of managing boundaries have presented similar situations in Japan and other countries. Therefore, the results of this study could be applicable to other countries. However, whether this study is applicable to other kinds of health-care facilities such as nursing homes or small hospitals, which have different organisational structures, will have to be considered in future studies.

Conclusion

Recently, hospital units have needed to operate more autonomously, and nurse managers have had to take greater responsibilities for managing their units' relationships with other units, professionals and departments. New perspectives on nursing leadership are also needed. Our findings provide preliminary but encouraging support for measuring boundary spanning roles of first-line managers. This study adds to our understanding of the managers' activities that indirectly support nursing practice. The results suggest that boundary spanning is one of the important roles of nurse managers and that further work is needed to refine this construct. Further research using the NMBS at other hospitals in diverse cities is needed to determine whether the same factors will be extracted and whether their reliability and validity can be verified. We believe that this instrument can offer health-care professionals and administrators guidance for improving nurse managers' behaviours, as well as for the effectiveness of nursing teams.

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Ethical approval

The study was approved by the University of Tokyo Graduate School of Midicine Research Ethics Committee (No. 3497).

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